

Manuscript type: **original paper**

Differences among lychee cultivars in responses to *Phytophthora litchii* nominate cell-wall and carbohydrate-associated candidates

Abstract: Litchi downy blight, caused by the oomycete *Phytophthora litchii*, affects young leaves, inflorescences and fruit of lychee (*Litchi chinensis*). We assembled five public sequencing cohorts for discovery, cross-context evaluation and auxiliary analyses of cultivar-dependent infection responses. Discovery in Guiwei and Yurong1 leaves tested 19445 expressed genes. Two hundred and six genes passed the DESeq2 discovery rule and mapping and gene-model quality control; 16 remained stable under alternative quantification, statistical analysis, expression filtering, leave-one-library-out analysis and mapping checks. Twelve were prioritised in Tier B, with provisional annotations related to sugar transport, starch metabolism, cell-wall remodelling, lipid signalling and RNA metabolism. In an independent pericarp time course, LITCHI005518, a putative alpha-xylosidase, and LITCHI001963 showed same-direction cultivar-dependent effects, whereas five genes reversed direction. Neither externally supported gene passed all internal robustness criteria. The frozen 206-gene signature showed tissue-dependent infection responses in Feizixiao. Most of 18 earlier candidates responded within at least one cultivar, but none passed the genome-wide cultivar-interaction criterion. These results distinguish general infection responses from cultivar-dependent effects and nominate context-qualified candidates for experimental validation.

Keywords: oomycete disease; host response; differential expression; transcript usage; reproducible bioinformatics; crop improvement

INTRODUCTION

Lychee (*Litchi chinensis*) is an important subtropical fruit crop grown widely in southern China and Southeast Asia. Litchi downy blight, caused by the oomycete *Phytophthora litchii*, attacks young leaves, inflorescences and fruit and contributes to serious field and postharvest losses. Cultivars differ in disease response, but the host processes associated with that variation remain incompletely resolved. Disease management still relies heavily on chemical protection, while resistance breeding requires candidates that can be tested as markers or functional determinants. The immediate biological problem is therefore to identify host genes whose infection responses differ among cultivars while separating them from genes that mark infection in all cultivars.

Previous studies provide several clues about the lychee response to *P. litchii*. A pericarp time course showed distinct early transcriptional programmes in susceptible and resistant cultivars (Sun et al. 2019), and the pathogen effector PIAvh202 promotes virulence by destabilising a host enzyme required for ethylene biosynthesis (Li et al. 2023). A pathogen transcriptome has also catalogued molecular machinery used during infection (Sun et al. 2017). ATP-regulated antioxidant and phenolic responses have been reported during litchi fruit pathogen infection (Yi et al. 2010), and LcCDPK expression changes under downy mildew stress (Liu et al. 2023). These studies establish biologically plausible defence pathways, but they differ in cultivar, tissue and sampling time. A gene highlighted in one experiment may therefore represent a general infection response, a cultivar-dependent response or a feature of the sampled tissue and time. Public RNA-sequencing datasets spanning cultivars, leaves and fruit, and several infection times make these possibilities testable within a common framework.

Our central question was which lychee genes change differently after *P. litchii* infection in different cultivars. This question is more specific than asking which genes change after infection. A gene strongly induced in both cultivars may participate in a general defence response but does not explain how cultivar responses differ. A cultivar-by-infection interaction instead tests whether the magnitude or direction of the infection response differs between cultivars. That distinction is important for prioritising candidates related to host variation and is statistically demanding because the available studies generally contain only three deposited libraries per cultivar and treatment group. Credible candidates therefore require genome-wide error control, checks for ambiguous read mapping and gene models, sensitivity analyses, and evaluation in independent biological contexts.

The project began with an exploratory reanalysis of the GSE201243 leaf dataset, which compared mock-inoculated and infected Guiwei and Yurong1 leaves 24 h after inoculation. Per-cultivar analyses nominated 18 genes with defence-related annotations and several stress- and hormone-associated promoter motifs. Because selection and characterisation occurred within one cohort, these findings established infection-response hypotheses but did not provide a genome-wide test of cultivar differences. We therefore treat this stage as exploratory and evaluate its candidates against the estimand they originally represented.

After the exploratory work, we registered a distinct secondary genome-wide analysis and fixed the roles of five public cohorts, the statistical models, decision thresholds and external evaluations before outcomes of the registered analysis were computed. We tested 19445

expressed genes for differences in infection response between Guiwei and Yurong1, assessed stability under alternative quantification and statistical procedures, and evaluated frozen candidates in independent pericarp and leaf-fruit datasets. The objectives were to identify quality-controlled cultivar-dependent responses, determine which signals survived internal perturbations, measure their transport across biological contexts, and integrate gene-level, pathway, transcript-usage, annotation and promoter evidence without allowing weaker evidence to replace expression support.

MATERIAL AND METHODS

Study design and registration. Registration followed exploratory analysis of the discovery data. Protocol version 1.0.0 was frozen on 18 August 2026 before outcomes of the registered secondary analysis were computed, and subsequent amendments were time-stamped and classified as outcome-blind or post-outcome technical changes. Five public cohorts had fixed, non-interchangeable roles (Table 1; Figure 1A). PRJNA830488/GSE201243 was the sole discovery cohort; PRJNA450886 was the primary cross-context evaluation; PRJNA922966/GSE222651 tested generic infection transfer; PRJNA922965/GSE222650 was assigned to small-RNA evaluation; and PRJNA1090613/GSE262200 remained exploratory because time and resistance metadata were unresolved. Discovery membership, signature weights and thresholds were frozen before external data were processed. We use cross-context support, rather than replication, for a passing external test because the external cohort differs in tissue, comparator or design. The amendment log records 20 outcome-blind clarifications, implementation corrections and optimisations made before discovery or external expression outcomes were viewed, plus three post-outcome amendments confined to technical quality control and manifest verification. The primary external study had been published and its broad conclusions were known before registration, but its count matrix was not processed until roles and thresholds had been fixed. The staged external unlock was therefore a workflow-enforced computational safeguard rather than experimental blinding. Frozen denotes a result fixed at a recorded checkpoint; robustness refers only to prespecified sensitivity analyses within discovery; and orthogonal evidence could not replace expression support.

Data provenance and eligibility. National Center for Biotechnology Information and Gene Expression Omnibus metadata for all five cohorts were reconciled into a biological-unit registry. The historical search log was not retained, so database coverage was reconstructed retrospectively and is not presented as prospective provenance. A super-series duplicate was

collapsed; PRJNA1157370 was excluded because it used *Colletotrichum gloeosporioides*, and PRJNA268587/GSE63658 was excluded because it examined fruit senescence and storage without *P. litchii* inoculation. The five prespecified cohorts therefore remained eligible. The primary discovery model required at least three included libraries in every cultivar-treatment cell. Because source-tree, pooling, harvest and extraction independence were not reported, inferential q-values remain conditional on the deposited libraries representing independent biological units; technical quality checks cannot establish that independence.

Preprocessing and quality control. The host SCAU_Lch_v2.0 assembly (Hu et al. 2022) and pathogen reference were combined with prefixed contig names to absorb pathogen-derived reads. Paired reads were checked, trimmed with fastp (Chen et al. 2018), aligned by STAR two-pass mapping (Dobin et al. 2013), counted with featureCounts (Liao et al. 2014) and independently quantified with Salmon (Patro et al. 2017). Libraries required at least 10 million surviving read pairs and 40% uniquely aligned reads. Exon uniqueness was measured with GenMap (Pockrandt et al. 2020), and candidate-level observed-mapping checks tested whether individual effects depended on ambiguously mapped reads. All 12 discovery libraries passed the prespecified library-level gates.

Genome-wide discovery. DESeq2 fitted gene counts as cultivar + treatment + cultivar:treatment, with Guiwei and mock as reference levels (Love et al. 2014). The primary interaction coefficient estimates the difference between the infected-minus-mock response in Yurong1 and that in Guiwei. Genes required counts of at least 10 in at least three libraries. Statistical candidates required genome-wide Benjamini-Hochberg $q < 0.05$ and an absolute estimated interaction log2 fold change of at least log2(1.5) (Benjamini & Hochberg 1995). The registered rule tested the point null of zero interaction and then filtered the observed estimate; it did not test whether the true effect exceeded that magnitude. Post hoc sensitivity analyses tested the composite null that the absolute effect did not exceed log2(1.5) and calculated apeglm false-sign-or-small values (Zhu et al. 2019); neither analysis altered frozen membership. Unshrunk maximum-likelihood estimates are reported, whereas shrunk effects were used only for signature weights.

Exploratory-candidate audit. To match the estimand used in the first-stage selection, separate infected-versus-mock DESeq2 models were fitted within Guiwei and Yurong1 using the same count matrix and 19 445-gene universe. Benjamini-Hochberg adjustment was performed genome-wide within each cultivar. These results were interpreted as within-cultivar infection

responsiveness and were kept distinct from the cultivar-by-infection interaction. The exploratory candidates were also subjected to the same exon-mappability and gene-model eligibility checks as genome-wide candidates. The first-stage exploratory pipeline used TopHat2/Bowtie2 alignment (Kim et al. 2013), HTSeq union-mode counting (Anders et al. 2015), and TBtools and PlantCARE for structural and promoter analyses (Chen et al. 2020; Lescot et al. 2002).

Internal robustness. Each frozen gene was retested using Salmon gene-level counts, edgeR quasi-likelihood interaction inference (Robinson et al. 2010), a counts-per-million expression filter and all leave-one-library-out refits. The Salmon gate required direction agreement and an absolute effect difference of no more than 0.5 log₂ units. edgeR used the same design and coefficient with robust dispersion estimation and required genome-wide $q < 0.10$ plus direction agreement. The leave-one-out gate required direction agreement in all 12 refits and $q < 0.10$ in at least 10. Gates were combined conjunctively, followed by observed-mapping sensitivity; no additive evidence score was used. edgeR applied TMM normalisation, robust dispersion estimation and quasi-likelihood testing, with Benjamini-Hochberg adjustment over the same 445-gene universe. The expression-filter analysis refitted the primary model after a counts-per-million filter rather than filtering a completed result. Pathway robustness likewise required agreement across competitive and rotation tests, leading-edge deletion and the matched-null comparison. A candidate passed only when every applicable requirement was met.

External evaluation and signed signature. PRJNA450886 was analysed as cultivar by treatment by time, with time categorical and 24 h the reference. The primary estimand was the Heiye-minus-Guiwei difference in infection response at 24 h; 6-h and 48-h contrasts were secondary. Candidate tests used Benjamini-Hochberg adjustment over the complete candidate-by-contrast family, an absolute log₂ fold-change threshold of log₂(1.5), a confidence interval excluding zero and discovery-direction agreement. Significant opposite-direction effects were recorded separately. PRJNA922966 was analysed as tissue + treatment + tissue:treatment for generic infection transfer within Feizixiao, not for between-cultivar effects. A frozen library-level signed score summarised the 206 genes, with shrunk discovery effects used as weights.

Pathways and transcript usage. Plant Reactome memberships were transported to lychee through one-to-one Oryza orthologues (Naithani et al. 2024). Discovery used the frozen interaction statistic, and pathway robustness combined competitive testing with camera (Wu & Smyth 2012), rotation testing with ROAST (Wu et al. 2010), leading-edge deletion and an

expression- and length-matched null. Transcript usage was tested from Salmon transcript abundances with DRIMSeq and DEXSeq under stageR stage-wise error control (Anders et al. 2012; Nowicka & Robinson 2016; Van den Berge et al. 2017). A registered conditional gate required interpretable gene- and transcript-level error control before event-level results could be reported.

Annotation and orthogonal evidence. Representative proteins were selected by longest coding sequence with lexicographic tie-breaking. Reviewed Swiss-Prot DIAMOND matches (The UniProt Consortium 2023), InterPro/Pfam domains and one-to-one Oryza orthology formed separate annotation classes (Buchfink et al. 2015; Jones et al. 2014). Promoter analysis tested 927 JASPAR plant motifs in strand-aware 1-kb and 2-kb windows against 100 expression- and GC-matched backgrounds, with AME as the primary enrichment test (McLeay & Bailey 2010; Rauluseviciute et al. 2024). Robustness required passing at least 80 backgrounds in both windows, and FIMO provided sensitivity testing (Grant et al. 2011). The PmiREN bulk archive was audited before small-RNA analysis (Guo et al. 2020); no study-derived reference was substituted if an exact species reference was absent.

Controlled promoter audit. A separate post hoc analysis examined whether the exploratory PlantCARE result depended on its background construction. The published totals for six reported element strings and one standard TCA-element variant were held fixed while the same scoring and multiplicity correction were applied to two backgrounds: 100 000 simulated sets of eighteen 2-kb sequences at 34% GC and 100 genomic sets matched by expression and promoter-GC quintiles. The randomized-GC rerun closely reproduced the exploratory pattern. Against matched genomic promoters, ARE lost significance, ABRE gained it, and MeJA-responsive, TCA and TC-rich classes remained significant. The comparison is provenance-limited because the original observed-motif input table was not archived and recounting the current canonical promoters reproduced the published total only for ARE. It was therefore treated as an audit of background sensitivity, not as evidence of regulatory function.

Power, evidence tiers and reproducibility. Negative-binomial simulations retained fitted means, dispersions, main effects and size factors while injecting balanced positive and negative interactions. Tier A required discovery, complete internal robustness, cross-context support, clean mapping and annotation, and attributable orthogonal evidence; Tier B required discovery and complete internal robustness with partial or non-testable external evidence; Tier C was reserved for a robust, externally supported pathway. Tier B denotes prioritisation, not complete

external validation. Snakemake workflows enforced staged analysis and SHA-256 manifests (Mölder et al. 2021). Complete tables, software versions, amendments and figure-generation checks are supplied in the supplementary archive.

Simulation design. Parametric power calculations used fitted gene means, maximum-a-posteriori dispersions, cultivar and treatment main effects, and library size factors from the discovery counts and the 24-h external subset. Equal numbers of positive and negative interactions were injected at 12 absolute log₂ fold changes from 0 to 4, with 100 simulations at each value. Discovery simulations injected 262 targets and applied genome-wide adjustment over 19 445 genes plus the registered effect filter. External simulations injected two targets among 177 frozen genes that passed count and dispersion validity checks and additionally required the prespecified direction and a 95% confidence interval excluding zero. Seven genes measurable in the observed 184-gene contrast were not fit-eligible for simulation. Dispersions were treated as known during simulated refits. The curves are conditional illustrations, not full-pipeline power estimates; the 24-h simulation should not be equated with the complete observed candidate-by-contrast testing family.

RESULTS

Discovery quality and genome-wide interaction. All 12 discovery libraries passed technical gates: read survival was 99.56-99.66%, unique alignment 84.44-86.22% and Salmon mapping 80.77-82.56% (Figure 1B). Principal component 1 separated cultivars and explained 54.2% of variance, while component 2 separated infection status and explained 16.1%; no library was isolated from its replicate group (Figure 1C). Among 19 445 expressed genes, 262 met both the registered genome-wide significance criterion and the observed-effect threshold. Uniform mappability removed 31 candidates and gene-model control removed 25, leaving 206 frozen candidates (Figure 2A,C). Recomputing the adjustment only among 13 602 QC-eligible genes recovered all 206 and added 12, yielding a larger set under a different testing universe. This comparison alone does not establish an FDR guarantee for post-test filtering. In post hoc sensitivity analysis, 8 of 262 statistical candidates and 7 of 206 QC-retained candidates passed the composite-null test; apeglm false-sign-or-small values were below 0.05 for 149 and 113 genes, respectively. These checks did not redefine the registered set, but limit claims that the true effects exceed the chosen magnitude.

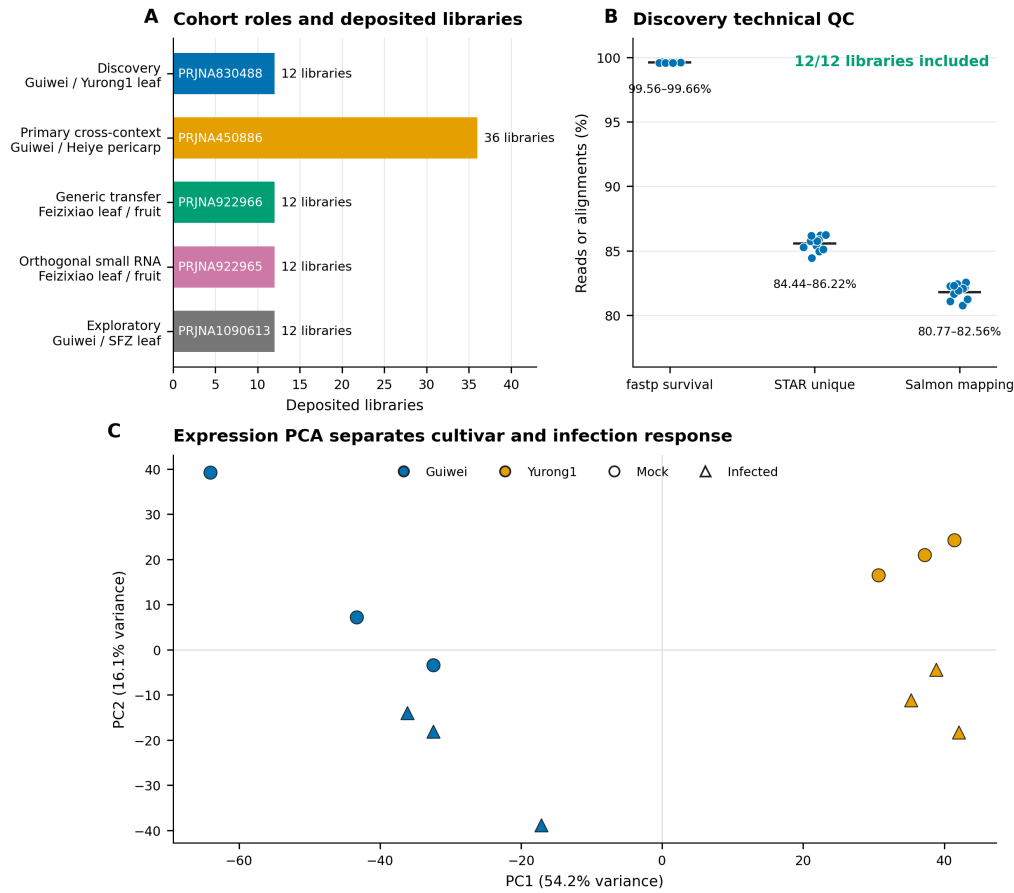


Figure 1. Study cohorts and discovery quality. (A) Five public cohorts with fixed roles. (B) Per-library read survival, STAR unique alignment and Salmon mapping; all 12 discovery libraries passed. (C) Principal components separate cultivar and infection status.

Table 1. Fixed dataset roles and eligibility.

Accession	Design	Fixed role
PRJNA830488 / GSE201243	Guiwei and Yurong1 leaf; 24 h	Discovery
PRJNA450886	Guiwei and Heiye pericarp; 6, 24, 48 h	Primary external
PRJNA922966 / GSE222651	Feizixiao leaf and fruit; 24 h	Generic transfer
PRJNA922965 / GSE222650	Feizixiao small RNA; 24 h	Orthogonal modality
PRJNA1090613 / GSE262200	Guiwei and SFZ leaf	Exploratory only

Exploratory candidates and internal robustness. The exploratory stage identified 17 responsive genes in Guiwei and 117 in Yurong1 and selected 18 defence-related genes. In estimand-aligned within-cultivar contrasts, 13 of 18 were significant in Guiwei and 16 of 18 in Yurong1 at genome-wide $q < 0.05$; 13 and 15, respectively, also had estimated effects exceeding the registered threshold. None met the separate cultivar-interaction criterion. LITCHI001510 reached interaction $q = 0.080$, and LITCHI019519 retained an interaction estimate of -3.33 with $q = 0.106$. LITCHI009301 and LITCHI028401 failed uniform mappability. Of the 206 frozen

genome-wide genes, 196 passed alternative quantification, 19 passed the edgeR gate, 205 passed expression filtering and 125 passed all leave-one-out refits. Eighteen passed these gates jointly and 16 remained after observed-mapping checks (Figure 3A). DESeq2 and edgeR estimates correlated at $r = 0.995$ genome-wide and $r = 0.999$ among candidates; all candidate directions agreed. Nevertheless, only 19 of 206 passed the edgeR significance gate. The significance discrepancy remains unresolved; close effect agreement does not establish cross-method robustness for the full discovery set.

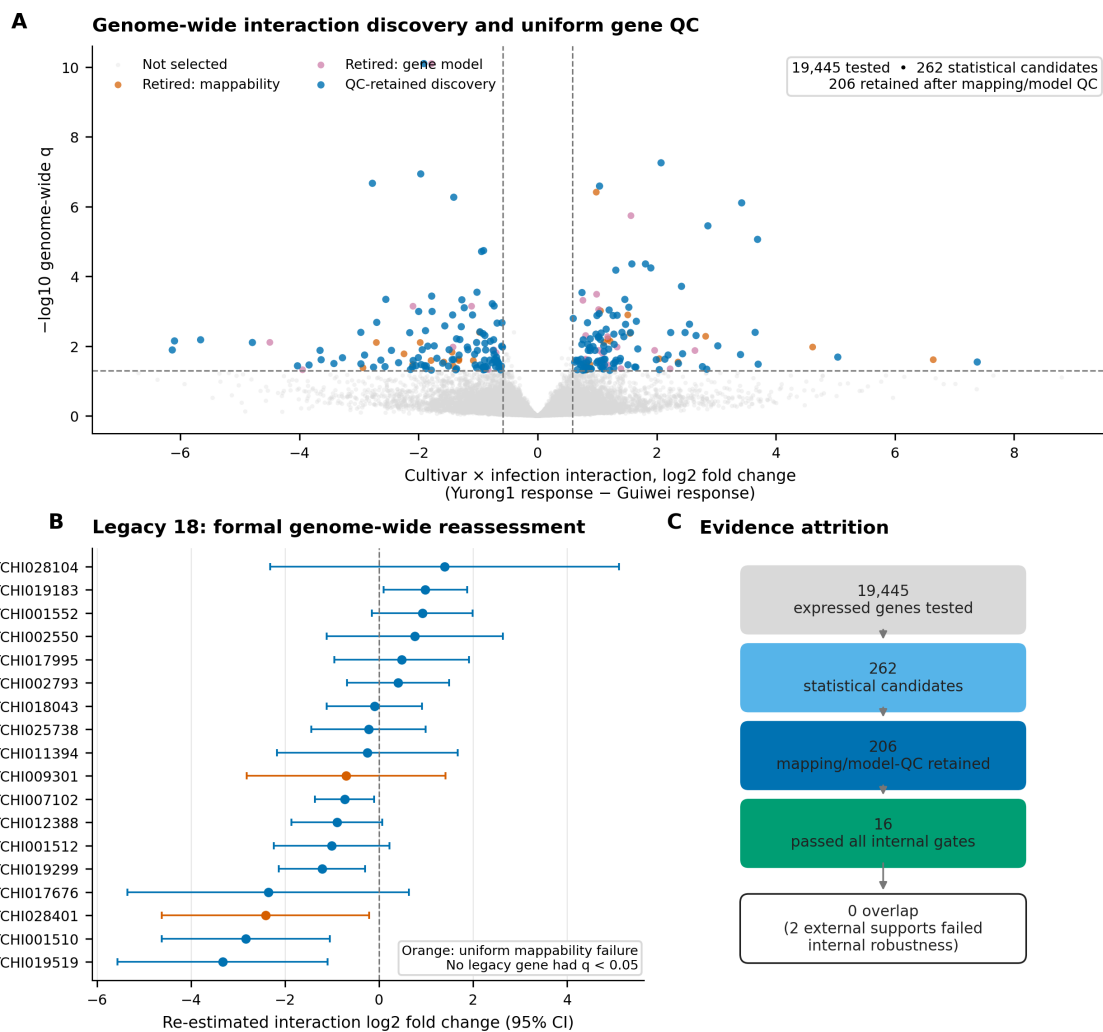


Figure 2. Genome-wide discovery and reassessment of exploratory candidates. (A) Interaction effects across 19 445 genes; 262 met the statistical threshold and 206 remained after quality control. (B) Interaction estimates for 18 exploratory candidates. (C) Attrition to the internally robust set.

Cross-context outcomes. In the primary 24-h pericarp evaluation, 184 of 206 candidates were measurable. LITCHI005518, provisionally annotated as an alpha-xylosidase, and LITCHI001963, an uncharacterised gene with a one-to-one *Oryza* orthologue, met every primary external criterion in the discovery direction (Table 2; Figure 3C). Five genes, including putative beta-glucosidase and Fe(II)/2-oxoglutarate-dependent dioxygenase genes, met

magnitude and significance criteria in the opposite direction. The remaining outcomes were below the support threshold or not testable. Across measurable genes, discovery and external effects showed little overall correlation (Pearson $r = -0.05$, 95% confidence interval -0.19 to 0.10; Spearman $\rho = -0.04$), and 95 of 184 shared direction. The 16 internally robust genes and two cross-context-supported genes did not overlap. An illustrative random-set calculation over all 206 candidates gives an expected overlap of 0.16 and a zero-overlap probability of 0.85; it is not a validation test (Figure 3B). Because tissue, cultivar comparator and study context differ together, the two positive outcomes provide cross-context support, not direct replication or evidence isolating tissue effects.

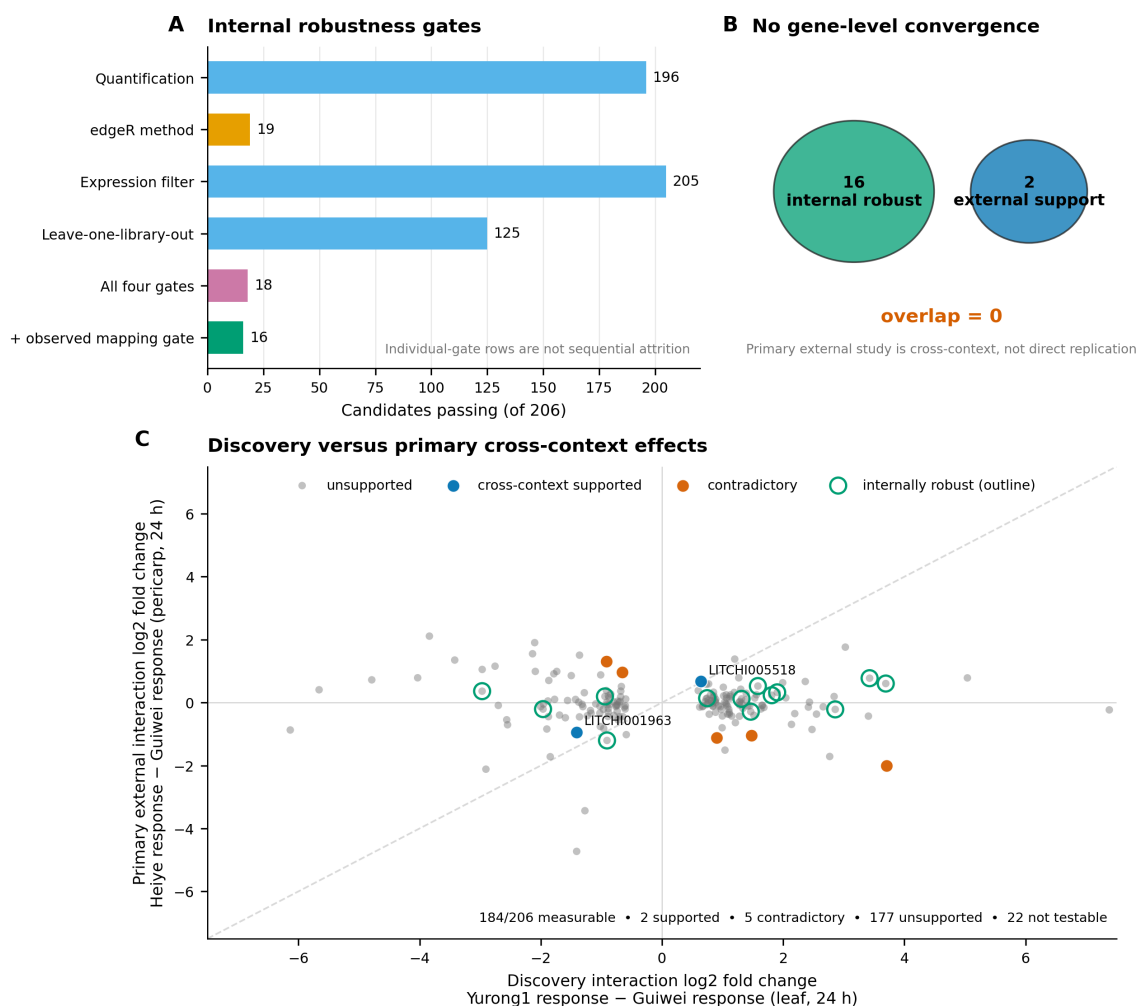


Figure 3. Internal robustness and primary external evaluation. (A) Retention by each robustness gate. (B) The 16 internally robust and two cross-context-supported genes do not overlap. (C) Discovery effects against 24-h pericarp effects; the diagonal is a visual reference. Support denotes the external criteria only.

Table 2. Supported and opposite-direction genes in the primary external evaluation. Effects are cultivar-by-infection interaction log2 fold changes; external intervals are 95% confidence intervals. Annotations are provisional. Neither supported gene passed every internal robustness gate.

Gene	Annotation	Discovery log2FC	Discovery q	External log2FC (95% CI)	External q	Outcome
LITCHI005518	Alpha-xylosidase 1	+0.64	0.025	+0.67 (+0.32 to +1.02)	0.007	Supported
LITCHI001963	Uncharacterised; 1:1 Oryza orthologue	-1.41	0.024	-0.94 (-1.43 to -0.46)	0.007	Supported
LITCHI030761	Caffeoylshikimate esterase	-0.92	0.040	+1.31 (+0.78 to +1.85)	0.0002	Reversed
LITCHI029534	Thiamine thiazole synthase 2	+1.48	0.002	-1.04 (-1.61 to -0.48)	0.010	Reversed
LITCHI021860	Phosphatidylinositol 4-kinase gamma 2	-0.65	0.032	+0.97 (+0.39 to +1.54)	0.029	Reversed
LITCHI008313	Beta-glucosidase 44	+0.90	0.045	-1.12 (-1.81 to -0.44)	0.034	Reversed
LITCHI001700	Fe(II)/2-OG-dependent dioxygenase 4	+3.70	0.033	-2.00 (-3.24 to -0.77)	0.035	Reversed

Signatures, pathways and transcript usage. The frozen 206-gene score tracked infection positively in Feizixiao leaf (0.107, $q = 0.0015$) and negatively in fruit (-0.181, $q < 0.001$), with a significant tissue interaction (0.289, $q < 0.001$). In pericarp, the cultivar-by-infection score effect was 0.109 at 24 h (95% confidence interval -0.079 to 0.297, $q = 0.349$) and -0.240 at 48 h (95% confidence interval -0.428 to -0.052, $q = 0.043$) (Figure 4B). These time-specific estimates suggest, but do not directly test, a difference between times. Six pathways were discovered, but none passed the primary external gates; circadian rhythm alone survived every internal pathway perturbation (Figure 4A). The conditional transcript-usage gate passed and identified 225 events across 152 genes among 15 790 tested transcripts. In the primary external cohort, 125 events were measurable but below support thresholds and 100 were not testable; annotation incompatibility prevented event-level testing in the generic-transfer cohort (Figure 5). In the exploratory cohort, four transcript-usage events met support criteria and five were contradictory, but these outcomes remained descriptive because time and resistance metadata could not be standardised. These transcript-usage candidates have discovery-stage support but no support meeting the prespecified external criteria; non-testable outcomes do not establish absence of an effect. The exploratory signed-signature estimate was likewise reported separately and did not contribute to any evidence tier.

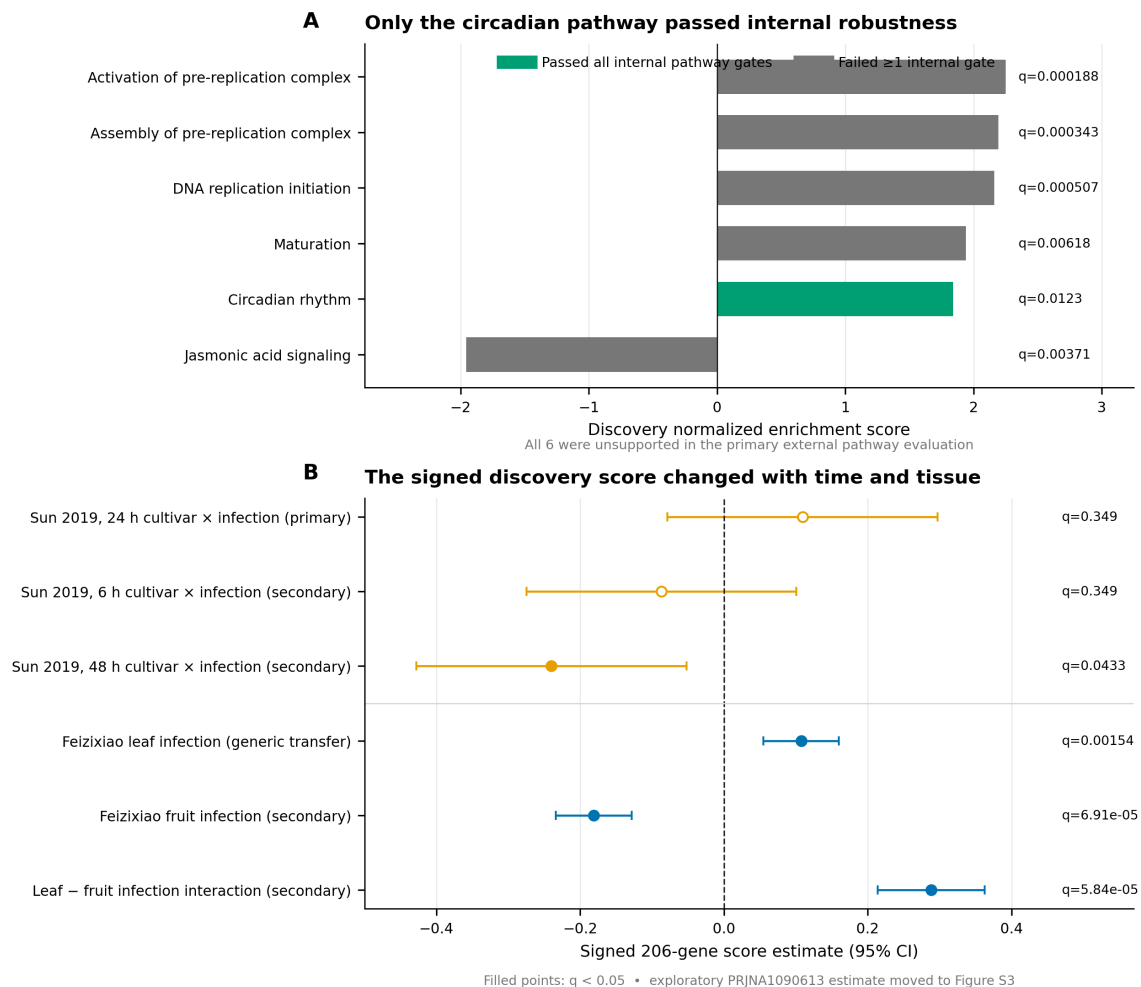


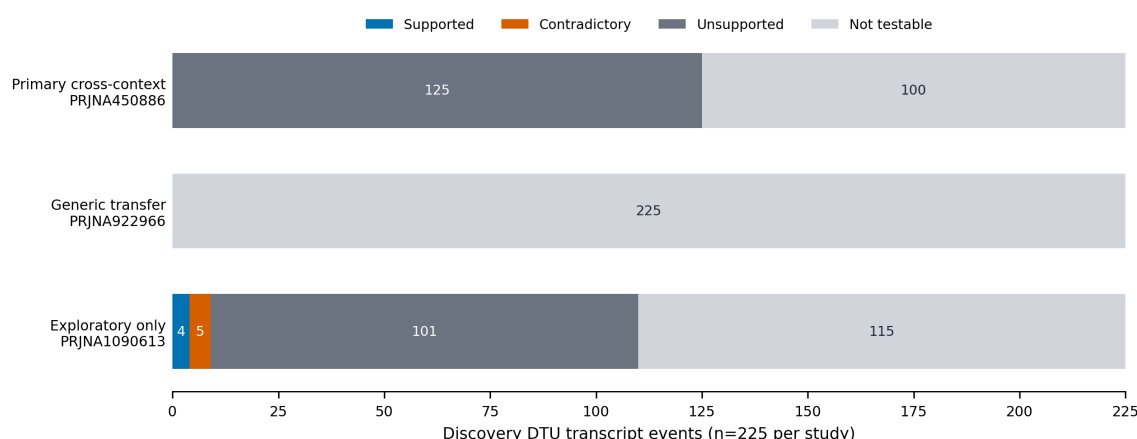
Figure 4. Pathway and signature evaluation. (A) Six discovery pathways; circadian rhythm passed all internal gates, but none passed external criteria. (B) The frozen 206-gene signature across six non-exploratory external contrasts; filled points denote $q < 0.05$. Tissue contrasts concern infection responses within Feizixiao; time-specific points do not test differences between times.

A Conditional DTU discovery



Stage-wise q values and signed usage-interaction differences

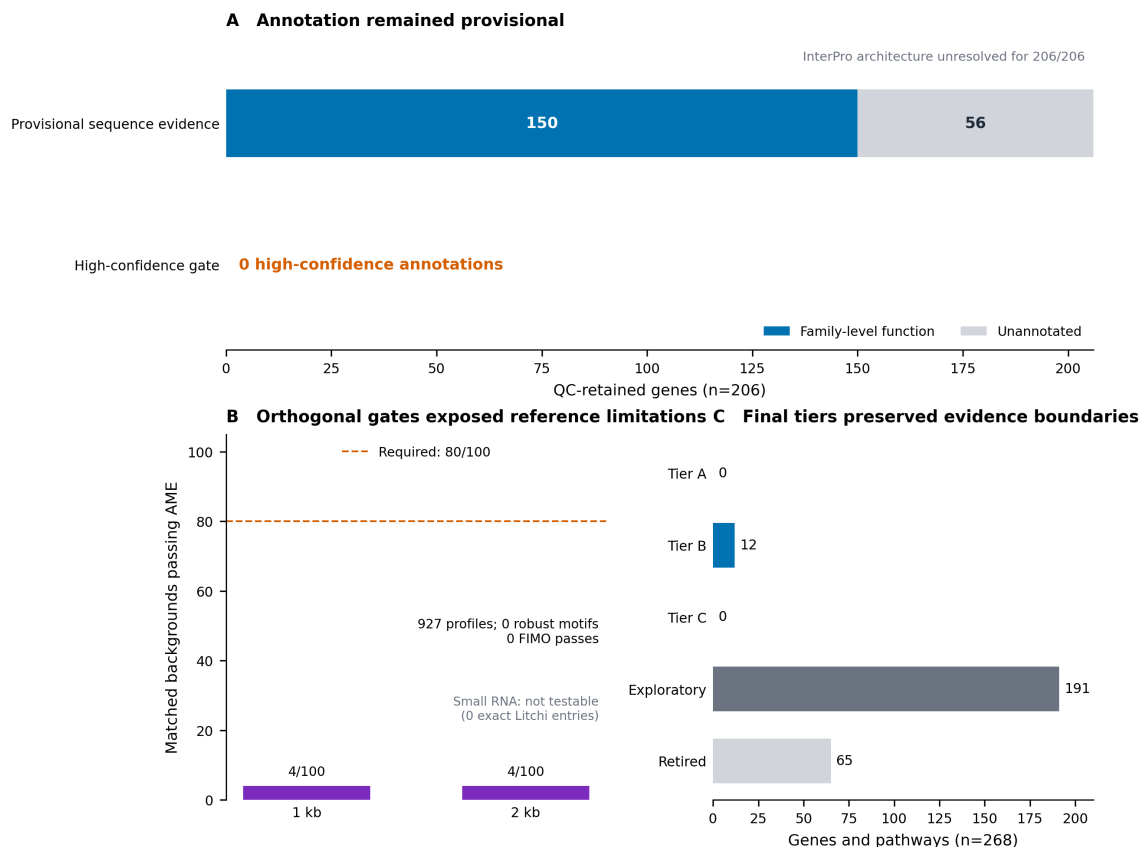
B External DTU follow-up remained separated by study role



No primary event was supported; exploratory outcomes remain non-confirmatory.

Figure 5. Conditional transcript usage. (A) Discovery gate and 225 events across 152 genes. (B) External follow-up separated by prespecified study role.

Orthogonal evidence, tiers and detectable effects. Provisional family-level labels were assigned to 150 candidates, but none satisfied the stricter requirement for two annotation classes, at least 70% supported coverage and no architecture conflict. No promoter motif passed the registered 80-of-100 matched-background rule in both windows, and small-RNA coherence was not testable because PmiREN contained no exact Litchi entry (Figure 6A,B). Evidence integration produced no Tier A gene or Tier C pathway, placed 12 internally robust genes in Tier B, retained 191 entities for exploratory follow-up and classified 65 as retired under distinct quality or external-direction rules (Figure 6C). Nine Tier B genes were measurable externally and all nine retained discovery direction without passing every support criterion; direction agreement alone is not external validation. Their annotations included carbohydrate and cell-wall metabolism, membrane and lipid transport, chaperone regulation, specialised metabolism, RNA metabolism and chromatin-associated regulation. Conditional simulations estimated 80% detectable effects of 2.44 log₂ units in discovery and 2.11 in external candidate-family analysis; at an injected absolute effect of 1.5 log₂ units, detection probabilities were 62.7% and 74.0%, respectively.



Tier composition and candidate functions. The 65 retired entities comprised 56 statistical candidates failing mapping or gene-model control, five genes with opposite-direction external effects and four candidates failing observed-mapping or annotation checks. Quality failures weaken locus-level inference; external reversals may reflect context specificity. Four internally robust genes, including a putative RPP8-like disease-resistance gene, remained exploratory and were measurable but below external support thresholds. Tier B does not establish an external-validation advantage over them. The two externally supported genes remained exploratory because they failed internal robustness requirements. The 12 Tier B annotations included an ERD6-like sugar transporter, a WAT1-related protein, a phosphomannomutase/phosphoglucomutase, a starch-branching enzyme, a serine carboxypeptidase, a DIR1-like lipid-transfer protein, a BAG-family chaperone regulator, two RNA-metabolism proteins, a chromatin-associated homeobox-DDT protein, a cytochrome P450 and one unannotated protein. These provisional labels are neither a formal enrichment result nor experimentally established functions.

Figure 6. Annotation, orthogonal evidence and prioritisation. (A) Annotation of 206 candidates. (B) Motif robustness and small-RNA reference gate. (C) Reported tiers over 268 frozen entities.

DISCUSSION

This analysis identifies an internally robust subset of cultivar-dependent transcriptional responses in the lychee-*P. litchii* pathosystem and organises those responses for experimental follow-up. Among 262 genome-wide statistical candidates, 206 passed mapping and gene-model quality control, 19 were significant under both statistical frameworks, and 16 passed the complete registered robustness procedure. Despite close effect agreement, significance was method dependent. Twelve internally robust genes were prioritised in Tier B. Two further genes met the primary external criteria, but neither belonged to the 16-gene internally robust set. Internal robustness and cross-context support therefore describe distinct properties; no gene satisfied both. The full 206-gene set remains a DESeq2-selected, quality-controlled candidate set rather than a uniformly robust or externally validated set.

The relationship between the exploratory and registered stages illustrates why the statistical model must match the biological question. Per-cultivar differential expression and a cultivar-by-infection interaction are different estimands. A gene can respond strongly in one or both cultivars without a precisely estimated difference between those responses. Most exploratory genes were supported as infection responsive within at least one cultivar, yet none passed the genome-wide interaction threshold. For example, LITCHI019519 retained a large interaction estimate but substantial uncertainty, and mapping control identified two exploratory genes whose locus-specific effects could not be resolved. Their absence from the interaction set therefore does not refute infection responsiveness; it limits claims about differences between cultivars.

Provisional annotations nominate cell-wall and carbohydrate dynamics for follow-up. Tier B includes putative phosphomannomutase/phosphoglucomutase, starch-branching enzyme, ERD6-like sugar transporter and serine carboxypeptidase genes. Externally supported LITCHI005518 has an alpha-xylosidase annotation, suggesting a possible role in xyloglucan remodelling. This qualitative pattern accords with earlier lychee work (Sun et al. 2019), but is not a replicated pathway result. Other labels suggest DIR1-like lipid transfer, BAG-family chaperone regulation and RNA metabolism (Berka et al. 2022). Jasmonic acid signalling lacked external support, and the promoter audit adds no regulatory confirmation. These are functional hypotheses, not established resistance mechanisms. Interaction signs alone do not distinguish stronger induction from weaker repression.

The multi-cohort results show limited cross-context transfer. LITCHI005518 and LITCHI001963 met the primary external criteria, whereas five genes reversed direction. Because tissue, cultivar comparator and study conditions change together, these outcomes do not isolate a tissue-specific cause. The Feizixiao leaf-fruit interaction supports tissue dependence of infection responses within that cultivar, not of between-cultivar differences. The pericarp estimates suggest temporal variation, but their separate significance statuses do not test a 24-h versus 48-h difference. Below-threshold results can reflect uncertainty as well as biological differences; reversals are distinct from technical mapping failures. The two externally supported genes merit follow-up alongside the internally robust set, with their incomplete internal robustness retained as a qualification.

The orthogonal analyses yielded below-threshold or non-testable results. No registered motif passed in more than four of 100 matched backgrounds, versus the required 80, and none passed the site-level sensitivity test. The controlled PlantCARE comparison showed that expression- and GC-matched backgrounds changed which element class passed. However, missing foreground provenance prevents a complete explanation of the exploratory and registered motif differences. Further motif recounting cannot establish regulation. Annotation and small-RNA analyses exposed separate reference limitations: complete domain architectures were unavailable for the full candidate set, and the frozen miRNA archive contained no exact Litchi entry. These gaps limit functional interpretation. Independently, the lack of overlap between complete internal robustness and primary external support already precludes Tier A.

Important limitations arise from the public designs. Each discovery cultivar-treatment cell contains three deposited libraries, whose independence at source-tree, pooling and extraction levels is unreported; q-values are conditional on that assumption. Technical quality control and leave-one-library-out checks cannot establish biological independence. Metadata remain incomplete, including inoculum form, leaf developmental stage and wounding status. All quantification uses one reference assembly, so residual mapping bias is possible. The external cohort changes tissue, cultivar comparator and study context, and no same-tissue, same-time independent cohort was available. Interactions identify transcriptional response differences, not resistance determinants. No candidate has been functionally perturbed. Conditional simulations place 80% detection at approximately 5.4-fold in discovery and 4.3-fold in external candidate-family testing, with poorer performance for low-expression genes. These fixed-dispersion illustrations cannot fully explain weak external support or distinguish an absent effect from a

context-specific response. Follow-up should use independently infected material from both discovery cultivars, verify prioritised expression patterns by qPCR, and test putative alpha-xylosidase and DIR1-like functions. Promoter hypotheses require reporter or chromatin-binding assays. Phenotype-linked validation is needed before causal or breeding claims.

CONCLUSION

This registered secondary analysis identifies 206 quality-controlled interaction candidates, 16 internally robust genes, 12 Tier B candidates and two genes with primary cross-context support. No gene met both complete internal robustness and primary external-support criteria. Provisional annotations nominate cell-wall and carbohydrate-associated candidates, not validated resistance mechanisms. Most earlier candidates remain supported as within-cultivar infection responses rather than cultivar differences. The results separate infection responsiveness, internal robustness and cross-context support, and provide a qualified basis for experimental follow-up.

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